



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION - SEMESTER I (2023/2024)
CSC1041 - PROGRAMMING LABORATORY I

Answer **ALL** Questions

Time Allowed: **Three** Hours

PART A - (C PROGRAMMING)

Create a folder named **CSC1041_End_S22XXX** in your H drive. Use the Sublime Text editor and save your C program as **S22XXXQ1.c** where **S22XXX** stands for your registration number. Extra marks will be awarded for using good programming practices.

1. [60 Marks]

You may store a maximum of **10** project assignment records using structures and arrays.

TechFlow Solutions is a software consultancy firm that manages projects for various clients. Your task is to develop a menu-driven C program to manage employee project assignments and calculate their monthly earnings based on project rates and hours worked.

Earnings Formula:

Base Earnings = Hourly Rate $\times$ Hours Worked

If Hours > Min Hours/Week: Bonus = (Extra Hours $\times$ Hourly Rate $\times$ Bonus Rate)

Total Earnings = Base Earnings + Bonus

Project Categories:

Project ID	Project Type	Hourly Rate (Rs.)	Min Hours/Week	Bonus Rate
1	Web Development	2500	20	1.2x
2	Mobile Apps	3200	25	1.3x
3	Data Analytics	3800	15	1.4x
4	ML Solutions	4500	30	1.5x

- Define a structure to store employee project assignment details. **[5 Marks]**
- Implement a function to register a new employee project assignment. The function should include the following operations. **[10 Marks]**

1. Prompt the user to input:
 - a. Employee Name
 - b. Employee ID (6-character code like **EMP901**)
 - c. Project ID
 - d. Hours Worked This Week
 - e. Experience Level (1=Junior, 2=Senior, 3=Lead)
2. Calculate earnings using the formula below, with an additional experience multiplier:
 - a. **Junior (1): No multiplier (1.0x)**
 - b. **Senior (2): 1.1x multiplier on total earnings**
 - c. **Lead (3): 1.25x multiplier on total earnings**
3. Display an assignment summary:

Example Output:

```
Assignment Registered Successfully
Employee ID: EMP901
Name: Zahid
Project: Mobile Apps
Hours Worked: 30 hours
Experience: Senior
Base Earnings: Rs. 96000.00
Bonus Earnings: Rs. 20800.00
Experience Bonus: Rs. 12848.00
Total Weekly Earnings: Rs. 129648.00
```

- c) Write a function that displays all project categories and calculates basic statistics. **[8 Marks]**

Example Output:

```
=== PROJECT CATEGORIES ===
```

ID	Project Type	Rate (Rs.)	Min Hrs	Bonus Rate
EMP901	Web Development	2500	20	1.2x
EMP803	Mobile Apps	3200	25	1.3x
EMP205	Data Analytics	3800	15	1.4x
EMP205	ML Solutions	4500	30	1.5x

```
Statistics:
```

```
Average Hourly Rate: Rs. 3500.00
```

```
Highest Bonus Project: ML Solutions (1.5x)
```

- d) Write a function that allows the user to search for an employee by entering the Employee ID. **[7 Marks]**

If found, display:

```
Employee Found:
Name: Zahid
Current Project: Mobile Apps
Current Hours: 30
Current Earnings: Rs. 129648.00
```

If not found, display:

No employee found with ID: EMP901

- e) Write a function that displays:
1. Summary Report **[8 Marks]**
 - a. Total number of active assignments
 - b. Total company payout for the week
 - c. Average earnings per employee
 - d. Highest earning employee
 2. Project Distribution Analysis **[7 Marks]**
 - a. Count of employees in each project category
 - b. Total hours allocated to each project type

Example Output:

```
=== FINANCIAL SUMMARY ===
Total Active Assignments: 5
Total Weekly Payout: Rs. 15900.00
Average Earnings: Rs. 97500.00
Top Earner: John Smith (EMP105) - Rs. 156750.00

=== PROJECT DISTRIBUTION ===
Web Development: 2 employees, 85 total hours
Mobile Apps: 1 employee, 35 total hours
Data Analytics: 3 employees, 28 total hours
```

- f) Write a function to display all the senior level employees who have been working for more than 25 hours. **[8 Marks]**

Example Output:

```
High-Performance Employees (Senior/Lead, >25 hours):
EMP101 - Sarah Ahmed - Mobile Apps - 35 hrs - Rs.
150,568.00
EMP105 - John Smith - AI/ML Solutions - 42 hrs - Rs.
156,750.00
Combined Revenue Contribution: Rs. 307,318.00
```

- g) Create the main menu as shown in the example below. The menu should be displayed repeatedly until the user chooses to exit. Call the correct function for each option and handle any wrong choices. **[Marks 7]**

Example Menu:

```
=== TECHFLOW EMPLOYEE MANAGEMENT ===
1. Register New Employee
2. Display Project Categories & Stats
3. Search Employee
4. Generate Reports
5. Show High-Performance Employees
6. Exit
Enter your choice:
```

PART B - (PYTHON PROGRAMMING)

Copy the shared text files into the newly created folder. Save your Python programs inside the folder as S22XXXQ1.py and S22XXXQ2.py for each question where S22XXX is your registration number.

2. [20 Marks]

You are tasked with building a **Category-Based Expense Tracker** for a student who wants better control over their finances. The tracker should record categorized expenses, ensure historical data is preserved, and provide meaningful insights.

Requirements:

1. Daily Entry Mode:

- Prompt the user for a **category name** (Food, Travel, Study, Others).
- Allow the user to **enter multiple expenses** under that category. Each expense should include a **description** and an **amount**.
- Store this data in a text file named after the **category and date** (e.g., Food_2025-06-20.txt).
- If a file for that category and date already exists, append the new items to it (do **not** overwrite).

2. After data entry is completed, display:

- Total spent in that category **on that day**.
- Item with the **lowest cost**.
- Number of items that cost between **Rs. 100 and Rs. 300** (inclusive)

3. Insight Mode:

Create a function **analyze_category(category_name)** that:

- Scans all files related to the given category (e.g., Travel_*.txt).
- Calculates and prints the:
 - Total number of entries made in that category.
 - Highest single-day expense in that category.
 - Average number of items per day.

Example Output:

```
Enter category: Travel
How many expenses? 3

Item 1 - Description: Bus fare
Item 1 - Amount: 120

Item 2 - Description: Snacks
Item 2 - Amount: 80

Item 3 - Description: Taxi
Item 3 - Amount: 600

Data saved to Travel_2025-06-13.txt

--- Summary for Travel (2025-06-13) ---
Total: Rs. 800
Cheapest item: Snacks (Rs. 80)
Items between Rs. 100 and Rs. 300: 1
```

3.[15 Marks]

A user is shopping online for groceries and adds items to their cart. For each item, the user provide:

- **Item name** (string)
- **Quantity** (a positive integer)
- **Price per item** (a positive float)

Design a Python program that:

1. Prompts the user to enter **details for 4 different grocery items**.
2. Validates the quantity and price (must be positive numbers).
3. Stores the data in a dictionary in the following format:

```
cart = {
    "Apples": {"quantity": 2, "price": 1.50},
    "Bread": {"quantity": 1, "price": 2.00},
    "Milk": {"quantity": 3, "price": 1.20},
    "Eggs": {"quantity": 2, "price": 2.50}
}
```

4. Ask the user to input their **total budget** (a positive float) and
- Calculates the total cost of all items ($\Sigma \text{quantity} \times \text{price}$).
 - Compares the total cost against the budget and displays:
 - "Within budget!" + remaining amount if affordable.
 - "Over budget!" + excess amount if not.
 - Prints a summary of all items and the final budget result.

Example Output:

Enter details for 4 grocery items:

Item 1 name: Apples
Quantity of Apples: 2
Price per Apples: 1.5

Item 2 name: Bread
Quantity of Bread: 1
Price per Bread: 2.0

Item 3 name: Milk
Quantity of Milk: 3
Price per Milk: 1.2

Item 4 name: Eggs
Quantity of Eggs: 2
Price per Eggs: 2.5

Enter your total budget: 15

--- Cart Summary ---

Apples: 2 * \$1.50 each -> \$3.00

Bread: 1 * \$2.00 each -> \$2.00

Milk: 3 * \$1.20 each -> \$3.60

Eggs: 2 * \$2.50 each -> \$5.00

Total cost: \$13.60

Within budget! Remaining: \$1.40